Operations Research 415
Part III: Markov Chains

Lecture 24: Workforce Planning

Presenter: Dr Philip Venter

Department of Industrial Engineering

Workforce Planning (pp. 950–951)

Consider an organisation whose members are classified at any time into one of s groups (labelled $1, \ldots, s$)

During each time period, a fraction p_{ij} of those who begin the period in group i, begin the next time period in group j

Also, during each time period, a fraction $p_{i,s+1}$ of all group i members leave the organisation

Let $\boldsymbol{P}$ be the $s \times (s+1)$ matrix whose (i,j)-th entry is p_{ij}

At the beginning of each time period the organisation hires H_i members in group i

Let N_i be the number of group i members at the start of period t

We call $\boldsymbol{N} = [N_1, N_2, \cdots, N_s]$ the steady-state census of the organisation

Workforce Planning (pp. 950–951)

For a steady-state census to exist, we must have that:

\# of people entering group i during each time period
    = # of people leaving group i during each time period

That is,

$$H_i + \sum_{k \neq i} N_k p_{ki} = N_i \underbrace{\sum_{k \neq i} p_{ik}}_{1 - p_{ii}}, \quad i = 1, \ldots, s \qquad (*)$$

- Given the values of the p_{ij}'s and the hiring policy (*i.e.* the values of the H_i's), the system (*) can be used to compute the steady-state census (*i.e.* the values of the N_i's)
- Given the values of the p_{ij}'s and the steady-state census (*i.e.* the values of the N_i's), the system (*) can be used to compute the corresponding hiring policy (*i.e.* the values of the H_i's)

Example 10, pp. 952–953 (Law firm example continued)

Let's return to the law firm of Mason and Burger
(see Example 8, pp. 946–947).

Suppose the firm's long-term goal is to employ 50 junior lawyers, 30 senior lawyers and 10 partners.

To achieve this steady state census, how many lawyers of each type should the law firm hire each year?

Additional example

The law firm of Mason and Burger hires 10 junior lawyers and 1 partner every year. They also fire 5 senior lawyers every year.

How many of each type of employee is expected to be at the firm?